

FRA INTELLIGENCE & DECISION SUPPORT PLATFORM

LOCKED PROJECT SCOPE

Hackathon build target: 12-hour implementation

CORE PRODUCT: Monitor -> Detect -> Investigate -> Explain -> Prioritize

The map is the interface; the real product is an evidence-backed anomaly and decision-support pipeline.

1. Product Definition

Locked product statement: An AI-powered FRA Intelligence & Decision Support Platform that helps administrators discover anomalous claim patterns, investigate the underlying evidence, and prioritize where human intervention is needed.

Not the product: a generic FRA dashboard, an FRA chatbot, or a map with red dots.

What makes it a product: Monitor the system, detect abnormal patterns, drill into evidence, generate an evidence-backed explanation, and rank cases for investigation.

2. Scope Principle

Everything in the build must strengthen one coherent workflow:

MONITOR -> DETECT -> INVESTIGATE -> EXPLAIN -> PRIORITIZE

3. MVP - MUST BE DONE

The MVP is intentionally constrained, but it is designed to be competition-worthy on its own. The full end-to-end investigation workflow must work without manual database intervention.

MVP 1 - National / State / District Intelligence Map

- Interactive GIS-style map with India -> State -> District drill-down.
- Map is the primary navigation surface, not decoration.
- Risk visualization using clear normal / watch / high-risk states or districts.
- Use official aggregate statistics where practical; use explicitly labelled synthetic claim-level data for detailed demo records.

MVP 2 - FRA Operational Dashboard

- District/state KPIs such as claims received, settled, pending, approval rate, median processing time and high-risk claim count.

- Status distribution and concise trend/summary visuals.
- Analytics should surface where abnormal behaviour is occurring rather than only report totals.

MVP 3 - Multi-factor Anomaly Engine

- Processing anomaly: compare processing duration with a local baseline/distribution; describe as statistically unusual, not legally overdue.
- Record inconsistency: compare fields such as claimed land area vs recorded area.
- Duplicate/similar claim detection using combinations such as village, land/survey reference, claimant identifier, area and location.
- Spatial anomaly: identify suspicious concentrations or geographic inconsistencies.
- Combine signals into an overall anomaly/risk score with explainable contributing factors.

MVP 4 - Claim Investigation Workspace

- Claim detail view showing identity, location, status and core attributes.
- Process timeline from submission through review/approval stages used in the demo model.
- Record comparison showing the exact values behind a mismatch.
- Claim location on the map.
- Visible "Why was this flagged?" evidence panel.

MVP 5 - Evidence-backed AI Investigator

- LLM receives structured evidence produced by the analytical pipeline.
- Generate human-readable assessment: findings, reasoning, confidence and limitations.
- AI must not invent the anomaly; it explains the evidence already established by the system.
- Explicitly surface unresolved questions / missing evidence instead of pretending certainty.

MVP 6 - Decision Prioritization

- Rank high-priority districts/claims by risk score.
- Show the dominant reasons driving priority.
- Provide an investigation recommendation, not an automated decision to approve/reject a claim.
- One-click path from priority queue to the underlying evidence.

4. Locked MVP Architecture

The technical architecture should keep the analytical truth outside the LLM and use the LLM mainly for interpretation and communication.

Layer	Responsibility	MVP output
Data	Normalize official aggregates + synthetic claim/event records	Clean state/district/claim dataset
Analytics	Processing, consistency, duplicate/similarity and spatial analysis	Anomaly signals
Risk engine	Combine signals into explainable risk score	Priority + contributing factors
GIS / dashboard	Map + KPIs + drill-down navigation	National -> state -> district views

Layer	Responsibility	MVP output
Investigation	Claim timeline, evidence and record comparison	Auditable case view
AI investigator	Summarize structured evidence and uncertainty	Evidence-backed narrative
Decision support	Rank cases and recommend where to investigate	Investigation queue

5. Data Strategy

Use a hybrid data strategy so the demo is credible, reproducible and controllable within 12 hours.

- Real/official aggregate FRA statistics: state-level totals and other published indicators used to ground the overview.
- Synthetic detailed records: district, claim, event, land and location data designed specifically for the demo.
- All synthetic/demo records must be explicitly labelled as demo/synthetic data; never present them as live government records.
- Design the synthetic dataset around known test scenarios rather than random noise.

6. Synthetic Demo Scenarios

Scenario	Signals	Purpose in demo
A - Processing bottleneck	Claim processing duration far above district baseline	Shows temporal anomaly
B - Land record mismatch	Claimed area materially differs from recorded area	Shows cross-record inconsistency
C - Duplicate-like claims	High similarity on identifiers/location/area	Shows entity/similarity reasoning
D - Spatial concentration	Unusual cluster of claims in a small area	Shows geographic analysis
E - Multi-signal hero case	Delay + mismatch + duplicate/spatial signal	Main 3-minute investigation case

7. Important Domain Guardrails

- Do not label a claim "legally overdue" based on an invented universal deadline. Use terms such as "statistically unusual processing time" or "workflow bottleneck".
- Do not let the AI declare a claim valid/invalid or instruct an automatic approval/rejection.
- Frame outputs as investigation prioritization and evidence review for authorized human decision-makers.
- The system should say when evidence is incomplete or conflicting.

8. POST-MVP - ONLY AFTER MVP IS STABLE

These features increase ceiling and polish, but none may block completion of the MVP.

POST-MVP 1 - Natural-language Investigation

Ask questions such as why a district is high-risk or which districts should be investigated first; route requests to the structured analytics/evidence layer.

POST-MVP 2 - Cross-district / State Benchmarking

Compare processing time, approval rate, pending ratio, mismatch rate and anomaly density between comparable units.

POST-MVP 3 - Anomaly Clustering

Group many alerts into recurring patterns such as record mismatches, processing bottlenecks or geographic clusters.

POST-MVP 4 - Explainable Risk Decomposition

Show how each signal contributes to the overall risk score instead of exposing only a single number.

POST-MVP 5 - Claim Relationship Graph

Visualize relationships between claims, land records, villages, survey references and locations for investigations.

POST-MVP 6 - Evidence / Source Drawer

Expose raw supporting records behind each finding, with quick compare/view actions.

POST-MVP 7 - Intervention Simulation

Optional what-if analysis such as projected backlog reduction under selected interventions; only attempt after all core flows are polished.

9. Explicitly Out of Scope

- Real government production integration or operational deployment.
- Full nationwide real-time claim-level ingestion.
- Complex authentication / RBAC beyond what the demo actually needs.
- Dedicated mobile application.
- Large model training or elaborate custom ML training pipelines.
- Real-time streaming infrastructure.
- Full digitization of every FRA process, record type and exception category.
- AI making final legal/administrative decisions.

10. Build Priority

Priority	Features	Rule
RED - MUST	Map; dashboard; anomaly engine; investigation page; evidence	Do not start post-MVP until these work

Priority	Features	Rule
	explanation; AI investigator; prioritization	end-to-end
ORANGE - POST	Natural language; benchmarking; clustering; risk decomposition; relationship graph; evidence drawer	Add only after a stable MVP demo
YELLOW - LAST	Intervention simulation	Only if everything else is polished

11. Locked 3-Minute Demo Story

0:00-0:30 - Problem: Introduce fragmented FRA monitoring and the need to identify where attention is most needed.

0:30-1:00 - National view: Show India map, key KPIs and the system identifying high-risk districts.

1:00-1:45 - Drill down: Open one district, reveal its unusual patterns and select the highest-priority claim.

1:45-2:20 - Investigate: Show timeline, record mismatch, spatial context and the exact evidence behind the anomaly score.

2:20-2:45 - Explain: Run the AI investigator and show the evidence-backed explanation plus uncertainty / unresolved evidence.

2:45-3:00 - Prioritize: Show the ranked investigation queue and close on the human-in-the-loop impact: faster, evidence-guided prioritization.

12. MVP Completion Gate

The MVP is considered complete only when the following can be demonstrated live without manually editing the database or hardcoding the displayed conclusion:

- Open application -> view national map.
- Click a state -> view district metrics.
- Open a high-risk district -> see anomaly indicators.
- Open a suspicious claim -> see timeline, records and location.
- View the exact reasons the claim was flagged.
- Run the AI investigator -> receive an evidence-backed explanation with limitations.
- Return to the priority queue -> see the case ranked for human investigation.

13. Why This MVP Is Designed to Win

Judging criterion	How the locked scope addresses it	Primary proof in demo
Innovation & Originality	GIS + anomaly intelligence + evidence-grounded AI + decision prioritization for FRA implementation	Unusual risk map -> explainable investigation
Technical Complexity	Data normalization, temporal/statistical/spatial analysis, similarity checks, risk scoring and AI	Multi-signal hero case

Judging criterion	How the locked scope addresses it	Primary proof in demo
	reasoning	
UI/UX & User Delight	Map-driven navigation, progressive drill-down, visual evidence and focused investigation workflow	India -> district -> claim -> explanation
Pitch & Demonstration	One coherent 3-minute story rather than a feature catalogue	Single high-risk district/claim narrative
Real-World Impact	Helps officials prioritize where to investigate rather than pretending to automate legal decisions	Ranked investigation queue + human-in-the-loop framing

14. Research Grounding / Scope Notes

This scope is grounded in the hackathon problem statement and checked against official FRA information sources to avoid inventing operational/legal assumptions.

- Hackathon PS-7 asks for a WebGIS-style view of mock FRA claim data, AI anomaly flags, and a state-wise decision-support panel; it explicitly permits self-generated mock GeoJSON for a 10-hour demo. (Origin Hackathon Problem Statements PDF.)
- Official Ministry of Tribal Affairs reporting provides state-wise FRA progress information and an FRA reporting/MIS portal, which supports using official aggregates as the grounded overview layer.
- The FRA process includes multiple institutional stages; the demo event timeline should therefore model workflow stages rather than pretending every case is a single-step transaction.
- Processing-time anomalies should be framed statistically/operationally rather than as universal legal deadline violations.

Reference sources consulted: Origin Hackathon Problem Statements PDF; Ministry of Tribal Affairs FRA page (<https://tribal.nic.in/fra.aspx>); Ministry of Tribal Affairs FRA Act & Rules publication (<https://tribal.nic.in/downloads/FRA/FRAActnRulesBook.pdf>); PIB material on FRA institutional process (<https://www.pib.gov.in/newsite/erelcontent.aspx?lang=2®=48&relid=38884>).